satellites must correct for this. They measure time with their own clocks, then slow that time down to the rate of time flow on Earth before transmitting it to our phones.

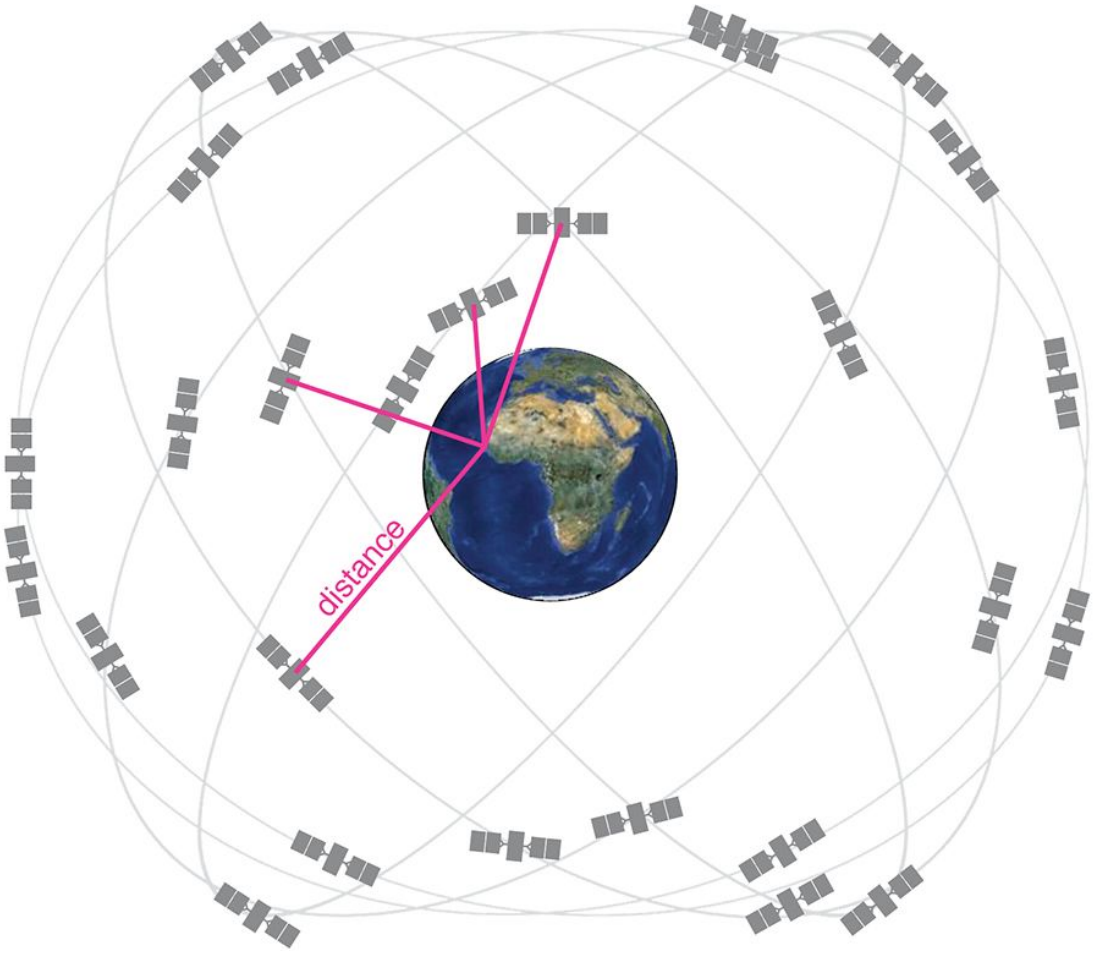


Fig. 4.2. The global positioning system.

Einstein was a genius. Perhaps the greatest scientist ever. This is one of many examples where his insights about the laws of physics could not be tested in his own day. It required a half century for technology to improve enough for a test with high precision, and another half century until the phenomena he described became part of everyday life. Among other examples are the laser, nuclear energy, and quantum cryptography.